

Amazon RDS (Relational Database Service) - Service Overview

Introduction

Amazon Relational Database Service (RDS) makes it easier to set up, operate, and scale a relational database in the cloud. RDS automates time-consuming administration tasks such as hardware provisioning, database setup, patching, and backups, while providing cost-efficient, resizable capacity.

Supported Engines

RDS supports several database engines:

- Amazon Aurora (MySQL and PostgreSQL compatible)
- PostgreSQL
- MySQL
- MariaDB
- Oracle Database
- Microsoft SQL Server

Multi-AZ Deployments

For production workloads, RDS offers Multi-AZ deployments, which automatically provision and maintain a synchronous standby replica in a different Availability Zone. In the event of infrastructure failure, RDS automatically fails over to the standby, typically within 60-120 seconds, without requiring manual intervention.

Read Replicas

Read replicas allow you to create one or more read-only copies of a database instance, using asynchronous replication. This offloads read traffic from the primary instance and can improve performance for read-heavy workloads. Read replicas can also be promoted to become standalone writable database instances if needed.

Storage

RDS uses Amazon EBS volumes for database and log storage. Storage types include General Purpose SSD (gp3), Provisioned IOPS SSD (io1/io2) for I/O-intensive workloads, and magnetic storage for backward compatibility. Storage can be scaled up without downtime on most modern instance classes, and Storage Auto Scaling can automatically increase allocated storage when RDS detects low free space.

Backups

RDS provides automated backups with a configurable retention period (1-35 days), enabling point-in-time recovery to any second within that window. Manual snapshots can also be taken at any time and are retained until explicitly deleted, independent of the automated backup retention period.

Security

RDS instances are typically deployed within a VPC, in private subnets not directly reachable from the internet. Access is controlled through VPC security groups, and data can be encrypted at rest using AWS KMS. IAM database authentication allows users to authenticate to certain engines (MySQL, PostgreSQL) using IAM credentials rather than a traditional database password.

Parameter Groups and Option Groups

Parameter groups act as a container for engine configuration values that are applied to one or more database instances. Option groups enable specific features, such as Oracle's Advanced Security option.

Amazon Aurora

Aurora is a MySQL- and PostgreSQL-compatible relational database engine built specifically for the cloud, offering up to 5x the throughput of standard MySQL and up to 3x the throughput of standard PostgreSQL on comparable hardware. Aurora separates compute from a distributed, self-healing storage layer that automatically replicates data six ways across three Availability Zones, and can scale storage automatically up to 128 TiB without any manual provisioning. Aurora Serverless additionally offers on-demand, auto-scaling compute capacity that scales up or down based on actual load, billed per second, well suited for applications with variable or intermittent database traffic.

Monitoring and Performance Insights

RDS integrates with CloudWatch to expose metrics such as CPUUtilization, FreeableMemory, and DatabaseConnections. RDS Performance Insights goes further, providing a dashboard of database load, broken down by SQL query, wait event, host, or user, making it straightforward to identify exactly which query is responsible for a spike in database load without needing to manually correlate slow query logs.

Common Use Cases

- Backing transactional web and mobile applications

- Storing structured metadata for applications where the primary search/analytics layer lives elsewhere (e.g. a vector database)
- E-commerce order and inventory systems
- SaaS multi-tenant applications requiring strong relational consistency

Best Practices

- Enable Multi-AZ for any production database where availability matters.
- Place RDS instances in private subnets, never expose them directly to the public internet.
- Enable automated backups with a retention period appropriate to your recovery point objective (RPO).
- Use read replicas to scale read-heavy workloads rather than vertically scaling the primary instance indefinitely.
- Rotate database credentials using AWS Secrets Manager rather than hardcoding them in application configuration.